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ORIGINAL ARTICLE

National and Regional Trends in Mental Health Disorders Among U.S. Adults Treated in the State Mental Health System

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Abstract

This study examined the national and regional temporal trends of major mental health disorders, including attention-deficit/hyperactivity disorder (ADHD), anxiety, bipolar, depressive, and trauma- and stressor-related disorders, among U.S. adults treated in the State Mental Health System from 2013 to 2022. All ten rounds of Mental Health Client-Level Data (MH-CLD), covering 37,944,795 adults aged 18 and older, were analyzed. Unadjusted and adjusted logistic regression models, controlling for age, sex, and race/ethnicity, were used to assess trends over time. The analyses revealed a significant increasing national trend in service-seeking patients with anxiety disorders (AOR = 2.38, 95% CI 1.76–3.23), trauma- and stressor-related disorders (AOR = 1.56, 95% CI 1.21–1.95), and ADHD (AOR = 1.54, 95% CI 1.36–1.76) between 2013 and 2022. In contrast, bipolar disorders showed a significant decline, from 25.99% to 15.19% (AOR = 0.56, 95% CI 0.48–0.66). Depressive disorder remained the most common diagnosis, but did not exhibit a significant national trend. Regional trends were generally consistent with national patterns, with some exceptions, such as a significant decrease in depressive disorders in the West and more rapid increases in anxiety disorders in the Midwest and West compared with national trends. Evidence from a large, ten-year cohort of patients reveals national and regional shifts, as well as persistent patterns, in the prevalence of adults seeking services for major mental health disorders in the U.S. State Mental Health System. These findings highlight the importance of national policies and the need for targeted, region-specific mental health strategies.

Keywords Mental health disorders · U.S. adult · National trends · Regional trends · Treatment · State mental health system

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Mental health integrates the dynamic interaction of cognitive, emotional, social, and behavioral processes that support an individual's ability to function and adapt across life stages (American Academy of Pediatrics, 2017; American Psychiatric Association, 2013; Kellogg, 2020). It exists along a continuum from wellness to illness and profoundly affects social participation, productivity, and overall well-being (Kessler et al., 1998; Scott et al., 2011; Wulsin & Singal, 2003). Globally, mental health disorders are a leading cause of disability, surpassing cardiovascular and cancer-related disease burdens in high-income countries (World Health Organization, 2022). An estimated one in eight individuals worldwide experiences a mental health disorder, with anxiety, depression, and trauma-related disorders accounting for most years lived with disability (Helaly et al., 2021; Kaushik et al., 2023).

In the U.S., nearly half of adults experience a mental health disorder at some point in their lives, and about one in four experience once annually (Kessler, 2005). The economic impact of major depressive disorder (MDD) alone exceeds 326 billion per year, reflecting both direct treatment costs and lost productivity (Greenberg et al., 2021). Mental disorders often co-occur with chronic conditions such as diabetes and cardiovascular disease, amplifying morbidity, mortality, and healthcare costs (Chapman et al., 2005; Parks et al., 2006). Individuals with both physical diseases and mental health disorders experience lower quality of life (Baumeister et al., 2011; Chong et al., 2012; Lim et al., 2012; O'Mahony et al., 2022) as well as shorter life expectancy (Momen et al., 2024). Despite these burdens, effective interventions remain underutilized and unevenly distributed across populations and regions (Institute of Medicine, 2006). Over the past decade in the U.S., diagnostic patterns for mental disorders have shifted. Diagnoses of anxiety, trauma-related disorders, and depressive disorder have shown steady increases, while bipolar and attention-deficit/hyperactivity disorder (ADHD) have stabilized or declined in some clinical settings, but not for all cohorts of population (Mojtabai & Olfson, 2024; Mojtabai et al., 2016).

Several frameworks help explain these divergent trends. The social determinants of mental health framework posit that socioeconomic and environmental conditions, including education, employment, housing stability, social support, neighborhood safety, discrimination, and access to healthcare, shape both risk exposure and resilience (Alegría et al., 2018; Allen et al., 2014; Lund et al., 2018). These determinants influence not only the onset and course of mental disorders but also help explain variations in prevalence and treatment utilization across populations. Individuals exposed to cumulative disadvantages such as poverty, chronic stress, and systemic inequities are more likely to experience heightened psychological vulnerability, while protective factors such as stable social networks, equitable healthcare access, and community cohesion can mitigate adverse outcomes(Giesbrecht et al., 2023; Pearlin et al., 1997; Reay, 2025) further suggests that chronic stressors accumulate and extend across multiple life domains, generating a compounding psychological burden over time. Recent global stressors, including the COVID-19 pandemic and political instability, and environmental uncertainty, have intensified this cumulative effects, particularly for anxiety and trauma-related disorders (Czeisler et al., 2020). Meanwhile, evolving diagnostic classification systems and insurance coding practices have reshaped how mental health conditions are identified and recorded (Clark et al., 2017). This process, often referred to as diagnostic substitution or diagnostic inflation, reflects a lowering of diagnostic thresholds and a reclassification of symptoms under broader categories, which may contribute to the apparent rise in anxiety and trauma-related diagnoses (Frances, 2013). These social and structural dynamics underscore how epidemiological trends in mental health reflect not only changes in population well-being but also shifts in diagnostic, policy, and health system contexts (Silverman et al., 2015).

Despite extensive literature on the prevalence and burden of mental health disorders, little is known about diagnostic trends, especially those served through the public State Mental Health System. Prior research has examined the drivers of these general population trends,

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but a critical gap exists in understanding how diagnostic trends evolve among adults actively receiving care within public mental health systems. Also, little is known about how diagnosis and service use differ across regions, which makes it difficult to understand how local resources and population needs affect care. This study fills that gap by examining national and regional trends over time in major disorders—ADHD, anxiety, bipolar, depressive, and trauma- and stressor-related disorders among U.S. adults treated in the State Mental Health System from 2013 to 2022.

Methods

Study Sample

The study used all 10 rounds of (2013–2022) of the Mental Health Client-Level Data (MH-CLD) dataset to analyze the temporal trend of different mental health disorders (SAMHSA, 2013-2022). The Substance Abuse and Mental Health Services Administration (SAMHSA) under U.S. Department of Health and Human Services (HHS) sponsored the MH-CLD administrative dataset. The MH-CLD framework collects demographic, clinical, and outcome data for individuals served by state mental health agencies (SMHAs) over a state-defined 12-month period. It provides a standardized way to track and analyze mental health service utilization and outcomes across states. The study's target population consists of participants who receive mental health and support services and are at least 18 years old. Every year since 2011, the MH-CLD has collected data on roughly 5 to 6 million patients who are undergoing treatment in outpatient, inpatient hospital, and residential programs. Additionally, between 2013 and 2022, MH-CLD gathered information on over 55 million participants. The study utilized approximately 3 to 4 million adults' data each year, sourced from the Mental Health Client-Level Data (MH-CLD) between 2013, the first year of publicly available data, and 2022, the most recent year available data. Due to the de-identified nature of this publicly available dataset, the study did not require Institutional Review Board (IRB) approval. The detailed study design and variables definition can be found SAMHSA website (SAMHSA, 2024).

Mental Health Disorders

The most recent mental health diagnoses that were active throughout the reporting year must be recorded by the states. For every client, up to three mental health diagnosis codes are reported, labeled as Mental Health Diagnosis One (MH1), Mental Health Diagnosis Two (MH2), and Mental Health Diagnosis Three (MH3). MH-CLD investigators used a modified version of the Clinical Classification Software (CCS) (AHRQ, 2016) to classify these codes into 13 distinct diagnostic categories and an additional "other" disorders group. A detailed list of ICD-9 and ICD-10 codes under each mental health category is available in SAMHSA's annual reports (SAMHSA, 2024). This study focused on five common mental health disorders—anxiety, depression, bipolar disorder, trauma- and stressor-related disorders, and ADHD due to their high prevalence. An adult was considered to have a disorder if it appeared as their first (MH1), second (MH2), or third (MH3) recorded diagnosis.

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Demographic Characteristics

MH-CHD collected sociodemographic characteristics of individuals, including age, sex, and race/ethnicity. For this study, age (18–24, 25–34, 35–44, 45–54, 55–64, and 65 +), sex (female, male), and race/ethnicity (Hispanic, non-Hispanic Black, non-Hispanic White, and Other) were included to examine temporal trends and for covariate adjustment. The "Other" racial/ethnic category included American Indian/Alaska Native, Asian, Native Hawaiian or Other Pacific Islander, and individuals identifying as some other race alone or more than one race.

Statistical Analysis

To assess the temporal trends from 2013 to 2022, the year variable was normalized to a 0-to-1 scale by subtracting 2013 from each year and dividing it by 9, allowing interpretation of changes over the full 10-year period. Initially, raw percentages were calculated for each year to illustrate the temporal trend of each disorder. Subsequently, separate logistic regression models were fitted with year as the predictor variable to examine these trends. Additionally, separate logistic regression models were fitted for each disorder for the entire U.S. and across regions (Northeast, Midwest, South, West) to examine both national and regional temporal trends after adjusting for covariates. When assessing the national-level trends, standard errors were adjusted for clustering at the regional level to account for regional variation within each disorder. Similarly, for regional trend analyses, standard errors were adjusted for clustering at the state level. All analyses were conducted in R (Version: 4.5.1) (R Core Team, 2024).

Results

Trends in Sample Characteristics

The distribution of the study sample from 2013 to 2022 by age, sex, and race/ethnicity is summarized in Table 1. Sex did not show a significant trend over time, though females consistently represented the majority (56.43% in 2013 and 55.86% in 2022; OR: 0.98, 95% CI: 0.90–1.07). More than 60% of patients were from the 18–34, 35–44, and 45–54 age groups. A significant decrease was observed in the 45–54 age group (OR: 0.65, 95% CI: 0.63–0.67), while significant increases were observed in the older age groups: 55–64 (OR: 1.13, 95% CI: 1.01–1.27) and 65 or older (OR: 1.57, 95% CI: 1.30–1.90). The majority of adults were non-Hispanic White (55.50% in 2013 and 51.81% in 2022), followed by Hispanic (23.56% in 2013 and 24.63% in 2022) and non-Hispanic Black (16.73% in 2013 and 16.91% in 2022). None of these groups exhibited a significant trend. However, the "Other" race showed a significant trend, rising from 4.21% in 2013 to 6.66% in 2022 (OR: 1.73, 95% CI: 1.11–2.70).

Trends in Mental Health Disorders

Table 2 presents the percentage distributions, unadjusted odds ratios, and adjusted odds ratios from the trend analyses for each disorder, both nationally and regionally. Significantly increasing trends were observed for anxiety disorders, ADHD, and trauma- and stressor-related disorders during the period. The anxiety disorder group exhibited the largest increase, rising from 16.02% in 2013 to 26.62% in 2022 (AOR: 2.38, 95% CI: 1.76–3.23), followed by the

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Table 1 Characteristics of adults included in the Mental Health Client-Level Data (MH-CLD) who received care in State Mental Health Systems between 2013 to 2022

Year	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	OR	95% CI	p-value
n	3,216,853	3,311,289	3,020,728	3,062,618	3,618,246	3,865,031	3,906,238	4,053,297	3,844,488	4,044,207			
Sex													
Female	56.47	55.93	55.55	55.82	55.46	55.39	55.50	55.09	55.75	55.86	0.98	0.90–1.07	0.634
Male	43.53	44.07	44.45	44.18	44.54	44.61	44.50	44.91	44.25	44.14	1.02	0.94–1.11	0.634
Age													
18–24	14.81	14.21	14.02	14.26	13.71	13.84	14.21	14.00	14.32	14.75	1.01	0.90–1.12	0.933
25–34	22.63	22.90	22.96	23.52	23.21	23.45	23.87	23.93	24.29	24.46	1.10	1.00–1.22	0.066
35–44	20.35	20.30	20.18	20.31	19.88	20.02	20.11	20.40	20.64	20.95	1.03	0.99–1.07	0.124
45–54	22.71	22.61	21.94	21.48	20.56	19.77	18.75	18.05	17.13	16.48	0.65	0.63–0.67	< .001
55–64	13.94	14.45	15.16	15.33	15.84	16.23	16.06	16.10	15.88	15.43	1.13	1.01–1.27	0.039
65 +	5.56	5.53	5.74	5.10	6.80	6.69	6.99	7.52	7.73	7.92	1.57	1.30–1.90	< .001
Race/Ethnicity													
Hispanic	23.56	25.41	25.70	23.61	18.58	20.35	20.16	21.64	24.68	24.63	0.95	0.34–2.64	0.915
Non-Hispanic Black	16.73	16.48	16.37	16.55	17.78	17.74	18.13	17.26	16.82	16.91	1.05	0.80–1.36	0.738
Non-Hispanic White	55.50	53.81	53.31	54.93	58.15	55.78	55.34	54.58	51.75	51.81	0.90	0.54–1.52	0.698
Other	4.21	4.31	4.62	4.91	5.49	6.13	6.37	6.51	6.75	6.66	1.73	1.11–2.70	0.015

OR = odds ratio, *CI* = confidence interval

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Table 2 National and regional trends in prevalence of different mental health disorders among adults receiving treatment in State Mental Health Systems between 2013 and 2022

Mental Health Disorders	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	OR	95% CI	p-value	AOR	95% CI	*p*-value
National																
n	3,216,853	3,311,289	3,020,728	3,062,618	3,618,246	3,865,031	3,906,238	4,053,297	3,844,488	4,044,207						
ADHD	3.47	3.51	3.57	3.84	4.08	4.27	4.13	4.08	4.53	4.67	1.52	1.36–1.70	<.001	1.54	1.36–1.76	<.001
Anxiety	16.02	16.85	18.24	19.67	21.64	22.92	23.68	24.12	25.70	26.62	2.30	1.68–3.15	<.001	2.38	1.76–3.23	<.001
Bipolar	25.99	25.37	23.42	21.43	19.00	17.84	16.79	16.50	15.57	15.19	0.55	0.46–0.66	<.001	0.56	0.48–0.66	<.001
Depressive	39.30	38.86	38.38	38.67	37.78	37.45	37.21	36.42	35.32	34.23	0.97	0.85–1.11	0.730	0.98	0.86–1.12	0.795
Trauma- and stressor-related disorders	15.22	15.41	16.39	16.39	17.50	17.51	18.19	18.86	18.87	19.29	1.54	1.24–1.90	<.001	1.56	1.21–1.95	<.001
REGION																
Northeast																
n	641,952	640,504	320,653	424,706	737,203	732,099	743,776	744,136	759,058	732,400						
ADHD	4.03	3.60	4.14	3.99	4.98	5.27	5.33	5.34	5.39	5.63	1.75	1.51–2.01	<.001	2.06	1.55–2.73	<.001
Anxiety	15.80	14.43	15.79	16.79	17.63	19.40	20.16	20.47	20.36	21.86	1.87	1.03–3.41	0.040	1.91	1.06–3.44	0.030
Bipolar	21.91	24.21	20.60	23.96	19.29	17.24	16.66	16.51	16.04	15.77	0.70	0.61–0.80	<.001	0.64	0.52–0.80	<.001
Depressive	39.72	40.70	36.13	40.28	39.52	38.78	38.22	37.77	37.28	35.99	1.07	0.93–1.23	0.352	1.11	0.97–1.27	0.128
Trauma- and stressor-related disorders	18.54	17.06	23.34	14.98	18.58	19.31	19.63	19.91	20.93	20.75	1.41	0.97–2.05	0.067	1.56	1.20–2.03	<.001
Midwest																
n	618,084	785,903	761,440	712,523	876,976	999,828	1,077,878	1,010,945	813,923	872,902						

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Table 2 (continued)

Mental Health Disorders	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	OR	95% CI	p-value	AOR	95% CI	*p*-value
ADHD	3.91	4.53	4.77	5.09	5.23	5.24	5.13	4.98	5.74	4.99	1.64	1.11–2.42	0.013	1.58	1.09–2.30	0.014
Anxiety	16.33	18.32	20.22	22.11	25.43	26.03	27.84	27.59	29.90	31.20	3.58	2.04–6.26	0.004	3.68	2.17–6.27	<.001
Bipolar	25.08	21.79	20.13	17.81	16.79	15.81	14.60	14.60	12.59	12.08	0.61	0.47–0.80	<.001	0.62	0.48–0.81	<.001
Depressive	39.77	39.33	38.27	38.64	35.77	35.98	35.11	34.68	34.32	33.01	1.19	0.95–1.48	0.135	1.19	0.96–1.48	0.111
Trauma- and stressor-related disorders	14.90	16.04	16.60	16.36	16.78	16.94	17.32	18.14	17.45	18.73	1.77	1.26–2.49	0.001	1.74	1.26–2.40	<.001
South																
n	1,179,739	1,132,949	1,141,813	1,098,761	1,166,079	1,194,496	1,186,811	1,241,879	1,190,450	1,397,646						
ADHD	3.05	2.94	3.07	3.15	3.25	3.32	3.30	3.04	3.57	3.78	1.30	1.12–1.50	<.001	1.29	1.11–1.51	0.001
Anxiety	15.79	16.33	17.03	18.28	18.76	20.26	20.12	20.35	22.39	22.49	1.76	1.24–2.50	0.002	1.84	1.31–2.61	<.001
Bipolar	29.35	29.21	27.11	23.96	22.35	21.31	20.71	20.44	19.98	19.44	0.59	0.46–0.75	<.001	0.61	0.48–0.77	<.001
Depressive	40.16	39.46	40.17	40.94	40.36	39.69	39.74	38.63	37.09	36.45	0.94	0.81–1.09	0.430	0.96	0.84–1.09	0.558
Trauma- and stressor-related disorders	11.65	12.06	12.62	13.67	15.28	15.42	16.12	17.54	16.97	17.84	1.80	1.45–2.24	<.001	1.89	1.52–2.35	<.001
West																
n	777,037	751,899	796,762	826,361	836,778	936,314	894,580	1,053,322	1,077,754	1,038,904						
ADHD	3.42	3.27	2.86	3.50	3.21	3.65	2.90	3.55	3.96	4.98	1.49	0.85–2.62	0.160	1.59	0.99–2.57	0.056
Anxiety	16.27	17.67	18.76	20.69	24.60	25.09	25.32	27.29	28.91	30.35	2.45	1.66–3.42	<.001	2.54	1.77–3.65	<.001

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Table 2 (continued)

Mental Health Disorders	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	OR	95% CI	p-value	AOR	95% CI	*p*-value
Bipolar	24.10	23.92	22.26	20.14	16.47	16.08	14.48	13.66	12.87	12.28	0.41	0.32–0.52	<.001	0.43	0.37–0.51	<.001
Depressive	37.34	36.18	36.55	34.69	34.95	35.30	35.89	34.74	32.94	31.40	0.85	0.75–0.96	0.009	0.84	0.76–0.94	0.001
Trauma- and stressor-related -	18.87	18.95	19.57	20.98	20.76	19.87	21.41	20.77	21.31	20.98	1.18	0.98–1.43	0.082	1.20	1.02–1.43	0.025

AOR = adjusted odds ratio; adjusted for age, sex, and race/ethnicity

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trauma- and stressor-related disorders, which increased from 15.22% in 2013 to 19.29% in 2022 (AOR: 1.54, 95% CI: 1.21–1.95), and ADHD, which rose from 3.47% in 2013 to 4.67% in 2022 (AOR: 1.52, 95% CI: 1.36–1.76, $p < .001$). In contrast, bipolar and depressive disorders showed declining trends, with the decrease in bipolar disorders being statistically significant. The bipolar disorder dropped by approximately two-fifths, from 25.99% in 2013 to 15.19% in 2022 (AOR: 0.56, 95% CI: 0.48–0.66). Although depressive disorder did not exhibit a significant trend, it consistently remained the most prevalent, decreasing slightly from 39.30% in 2013 to 34.23% in 2022 (AOR: 0.98, 95% CI: 0.86–1.12).

Across all mental health disorders examined, regional trends aligned with national patterns in terms of direction and statistical significance, except for depressive disorder. Notably, depressive disorder showed a significant decline in the West, despite exhibiting no significant change at the national level. Over the past decade, bipolar disorders exhibited a consistent decline across all regions, with the Western region demonstrating the most pronounced decrease. In contrast, anxiety disorders in both the Midwest and West regions displayed significantly higher increasing trends compared to the national average. Similarly, apart from the South region, ADHD showed more rapid increases across all regions relative to national trends.

Figure 1 illustrates the trends in diagnostic groups across patient age categories during the study period. The majority of diagnostic groups showed statistically significant trends across age categories, except for the depressive disorder and ADHD groups, which did not show statistical significance. During the study period, the prevalence of anxiety, bipolar, and depressive disorders was most notably elevated in the 18–24 age category, followed by the ADHD and trauma- and stressor-related disorders groups in the 45–54 age category.

Figure 2 presents the prevalence of mental health disorders by sex during the 2013–2022 study period. Except for depressive disorders, all trends were statistically significant. The prevalence of anxiety disorders was higher in female patients compared to male patients,

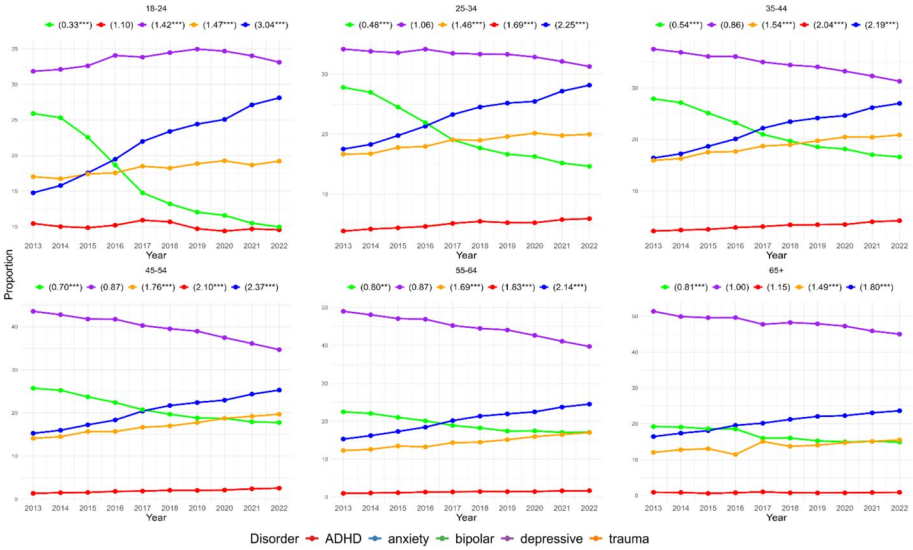

Fig. 1 Trends in prevalence of major mental health disorders among adults receiving treatment in State Mental Health Systems from 2013 to 2022, stratified by age group. Panels show annual proportions of adults diagnosed with attention-deficit/hyperactivity disorder (ADHD), anxiety, bipolar, depressive, and trauma- and stressor-related disorders. Color legends present the odds ratio (ORs) measuring changes over the study period and p-values: ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$

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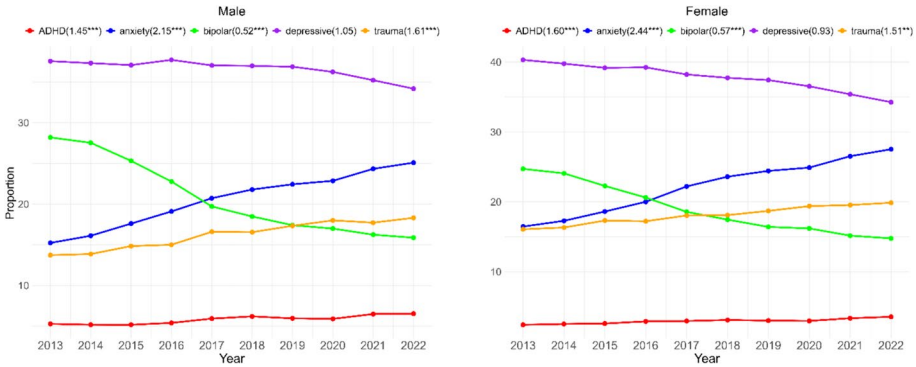

Fig. 2 Trends in prevalence of major mental health disorders among adults receiving treatment in State Mental Health Systems from 2013 to 2022, stratified by sex. Panels compare male and female adults, showing annual proportions diagnosed with attention-deficit/hyperactivity disorder (ADHD), anxiety, bipolar disorder, depressive disorder, and trauma- and stressor-related disorders. Color legends present the odds ratios (ORs) measuring changes over the study period and p-values: ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$)

and a similar pattern was shown for the ADHD category. In contrast, the prevalence of trauma- and stressor-related disorders was higher in male patients than in female patients, while female patients exhibited a higher prevalence of bipolar disorders compared to male patients over the study period.

Figure 3 illustrates the prevalence of mental health disorders by race/ethnicity over the study period. With the exception of depressive disorders, all trends were statistically significant. Across groups, anxiety and trauma-related disorders increased consistently, though the magnitude differed: Non-Hispanic White and "Other" groups experienced the largest increases in anxiety, while Hispanic and Non-Hispanic Black adults showed more gradual rises. Bipolar disorder declined steadily in every group, with the most pronounced decreases among Non-Hispanic White and Hispanic adults. Although depressive disorders remained the most prevalent diagnosis across all racial and ethnic groups, their rates were relatively stable over time, underscoring persistent unmet mental health needs even as other diagnostic patterns shifted.

Discussion

This study investigated the temporal trends in the prevalence of five major mental health disorders, including ADHD, anxiety, bipolar, depressive, and trauma- and stressor-related disorder, among U.S. adults treated in the State Mental Health System and analyzed their regional variations from 2013 to 2022. ADHD, anxiety disorders, trauma- and stressor-related disorders increased over time, while bipolar disorders declined. However, depressive disorder did not show a statistically significant trend. Across the regions, anxiety, ADHD, and trauma- and stressor-related disorders showed significant upward trends over time, while depressive disorders showed no significant trends in most regions, except in the West.

The incidence of anxiety disorders among U.S. adults has exhibited a widespread and consistent increase across geographic regions, aligning with a previous study that reported an approximate 7% rise in anxiety disorders from 2008 to 2018. By age group, young

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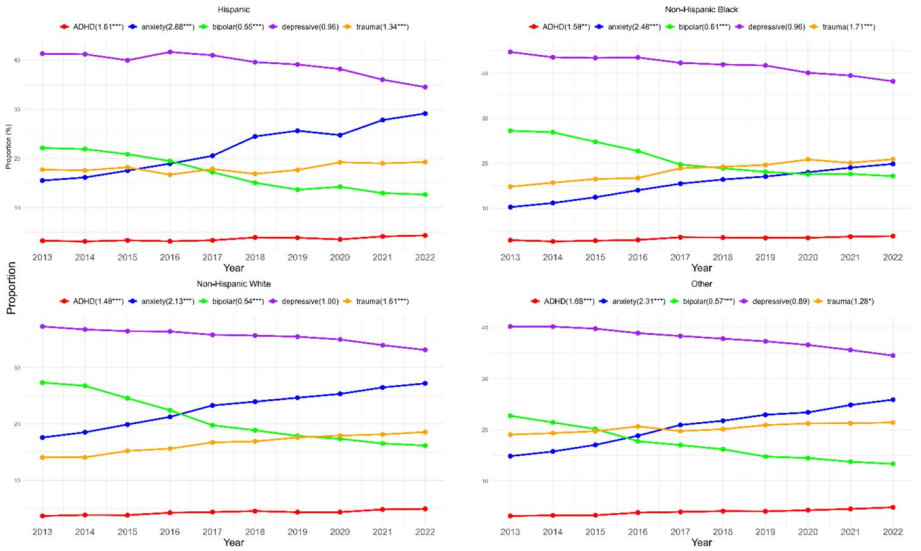

Fig. 3 Trends in prevalence of major mental health disorders among adults receiving treatment in State Mental Health Systems from 2013 to 2022, stratified by race/ethnicity. Panels show annual proportions of adults diagnosed with attention-deficit/hyperactivity disorder (ADHD), anxiety, bipolar disorder, depressive disorder, and trauma- and stressor-related disorders for Hispanic, non-Hispanic Black, non-Hispanic White, and Other racial/ethnic groups. Color legends show the odds ratios (ORs) measuring changes over the study period and p-values:***$p < 0.001$, **$p < 0.01$, *$p < 0.05$)

adults aged 18–24 years had the highest rates of anxiety disorders, consistent with earlier research, indicating the most rapid increase in this age group. This elevated prevalence requires further discussion, as it may be linked to factors uniquely affecting this demographic. For example, problematic engagement with social media, a behavior highly prevalent in this age group, has a significant positive association with anxiety symptoms (Shannon, 2022). A study in the United States reported that a significant increase in the incidence of anxiety disorders among females compared to males occurred between 2019 and 2022 (Terlizzi & Zablotsky, 2024). This finding aligns with our results, which showed a higher prevalence of anxiety disorders among females than males from 2013 to 2022, which further supports prior evidence noted in the introduction that anxiety disorders represent one of the fastest-growing mental health conditions in recent decades (Kessler et al., 2005; World Health Organization, 2023). The significant increase in anxiety and trauma-related disorders across racial and ethnic groups may reflect rising exposure to socioeconomic stressors and improved detection in ambulatory settings (Goodwin et al., 2020; Kavelaars et al., 2023). Some studies suggest that a range of socioeconomic and environmental factors, such as education, employment, housing stability, social support, neighborhood conditions, discrimination, and healthcare access, collectively influence individuals' vulnerability to mental health risks and their capacity for resilience. (Alegría et al., 2018; Allen et al., 2014; Lund et al., 2018).

Although our study found a relatively stable trend in diagnosed depressive disorders among all treatment-seeking U.S. adults, young adults showed a steady increase. This findings reflect the evidence of rising depressive disorders among young adult in U.S. from 2015 to 2020 (Goodwin et al., 2022) and global trends reported from 1990 to 2021 (Yang et al., 2024). Unlike older populations, younger adults may be more vulnerable to

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unique modern stressors such as economic instability, academic and occupational pressures, and the pervasive influence of digital technology and social media on psychological well-being (Khalaf et al., 2023). Furthermore, collective traumas like the COVID-19 pandemic have disproportionately affected younger age groups, contributing to elevated stress, social isolation, and depressive symptoms (Kleine et al., 2023; Shazley et al., 2025). This divergence across age groups underscores the complex interplay between sociodemographic factors, healthcare access, and evolving psychosocial environments in shaping depressive disorder trends. These finding align with the stress-vulnerability framework introduced earlier (Alegría et al., 2018) which emphasizes that biological, social, and environmental stressors interact to produce differential mental health outcomes across populations.

Several factors may explain the most common but stable depressive disorders over the years. First, even as screening increases, many new cases identified through tools like the 9-item Patient Health Questionnaire (PHQ-9) may represent milder or transient episodes rather than chronic major depression and over time, the pool of undiagnosed but more severe cases may shrink, limiting further growth in diagnoses (Patten & Schopflocher, 2009). Second, systematic depression screening has expanded, but one large health-system study found that routine PHQ-9 screening increased diagnosis rates only modestly, suggesting that higher detection rather than a true rise in incidence largely explained the change (Pfoh et al., 2020). Additionally, the system may have reached a diagnostic saturation point, where most individuals with recurrent or chronic depression are already engaged in care, resulting in a plateau in new cases recorded within the public system. Moreover, improvements in healthcare accessibility, early screening, and evidence-based interventions may have helped stabilize prevalence (Hu et al., 2022). Increased awareness of mental health issues among clinicians, alongside enhanced social support services and community-based programs targeting older adults, may also contribute to the plateauing of depressive disorders in recent years (Reynolds et al., 2022; WHO, 2023).

Our study revealed a decreasing trend in the prevalence of bipolar disorder among treatment-seeking U.S. adults from 2013 to 2022. Several factors may help explain this pattern. First, the adoption of stricter diagnostic criteria and more rigorous differentiation between bipolar and major depressive disorders in clinical practice have likely reduced over-diagnosis of bipolar disorders (Kessing et al., 2021). Second, persistent under-recognition of hypomanic or manic symptoms in primary care and community mental health settings may result in underdiagnosis or misclassification of bipolar disorder as unipolar depression (Shen et al., 2018; Singh & Rajput, 2006; Wu et al., 2024). Changes in help-seeking behaviors, treatment accessibility, increased reliance on outpatient management, and diagnostic practices across healthcare providers could also contribute to this downward trend.

During the same period, the prevalence of ADHD among U.S. adults rose from 3.47% in 2013 to 4.67% in 2022. This upward trend persisted into 2023, as reported in the Morbidity and Mortality Weekly Report (MMWR) by the U.S. Department of Health and Human Services (Staley et al., 2024). This increase may be linked to greater awareness among physicians and the public, along with the broadening of ADHD diagnostic criteria over time (Abdelnour et al., 2022). Similarly, a study in Japan also found increasing ADHD prevalence between 2010 and 2019 (Sasayama et al., 2022). Trauma- and stressor-related disorders among U.S. adults exhibited statistically significant increasing trends across all U.S. regions from 2013 to 2022. This upward trend mirrored rising stress levels linked to collective traumas as the COVID-19 pandemic. The 2023 *Stress in America* survey reported that 24% of U.S. adults experienced high stress (scores of 8–10 on a 10-point scale), up from 19% in 2019 (APA, 2023).

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Regional disparities were evident in diagnostic trends between 2013 and 2022, underscoring the influence of geographic context on mental health outcomes. The Midwest and West showed steeper increases in anxiety, ADHD, and trauma- and stressor-related disorders, while depressive disorders declined only in the West. These patterns may reflect demographic and structural factors such as higher rural population density, greater shortages of mental health providers, and persistent barriers to specialty care access, all of which can amplify unmet need and contribute to delayed or crisis-driven service use (Kim et al., 2017; Weeks et al., 2023). Policy variation also plays an important role. States differ widely in Medicaid expansion decisions, enforcement of mental health parity laws, and investment in behavioral health infrastructure, producing substantial regional variation in service availability and diagnostic patterns across regions (Pillai et al., 2024). Beyond system-level factors, regional differences in diagnostic culture, public awareness, health-seeking behaviors, and community stigma may influence how and when individuals engage with the mental health system.

This study has several notable strengths. A key strength is the use of nationally representative MH-CLD from 2013 to 2022, which included a large sample of U.S. adults, ensuring robust statistical power to detect temporal trends. The longitudinal design allowed further identification of national and regional increasing trend in anxiety, ADHD, trauma- and stressor-related disorders, providing critical evidence to guide policymakers in prioritizing early intervention, expanding access to services, and promoting public awareness campaigns across diverse geographic regions. Despite having strengths, this study has several limitations. First, the reliance on administrative state-level data consisting of clinician diagnoses, rather than standardized research diagnoses, may introduce heterogeneity due to varying clinical practices and lack of reliability testing. This variability could lead to inconsistencies in how mental health disorders are identified and recorded, potentially affecting the accuracy and comparability of trends. Additionally, these administrative datasets are susceptible to reporting bias, which may be influenced by clinician preferences, institutional policies, or evolving coding practices over time. Second, the dataset includes limited sociodemographic information, as key variables such as residential ZIP code, household structure, education level, employment status, income, insurance coverage, housing stability, neighborhood deprivation, exposure to violence or discrimination,, and broader socioeconomic, contextual, and environmental factors (e.g., air quality, green space access, urbanicity, and regional healthcare availability) are unavailable, constraining contextual interpretation. Third, generalizability is also limited, as the data predominantly reflect treatment-seeking populations, excluding untreated or undiagnosed cases, potentially leading to an underestimation of prevalence and trends in the broader U.S. population. Fourth, while the study provides evidence of temporal trends in mental health disorders among U.S. adults from 2013 to 2022, the dataset does not allow identification of the specific reasons underlying changes in mental health diagnoses. Finally, the potential confounding effects of the COVID-19 pandemic, including disruptions in healthcare access, changes in healthcare-seeking behavior, increased adoption of health, heightened awareness and screening for mental health issues, and shifts in diagnostic or reporting practices, cannot be fully disentangled from the observed diagnostic patterns.

In conclusion, the observed trends in mental health disorders among U.S. adults highlight the urgent need for targeted public health strategies and continued research. The consistent rise in anxiety, ADHD, and trauma disorders across all U.S. regions underscores the importance of expanding timely access to mental health services, particularly for young adults (18–24 years), and strengthening the clinical workforce to meet evolving needs. These patterns carry significant implications for mental health professionals and service

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delivery systems, indicating the need to strengthen workforce capacity, enhance training in evidence-based and trauma-informed care, and adopt culturally responsive approaches to meet diverse population needs. To address these growing challenges, key system-level interventions include strengthening community-based services, integrating routine mental health screening into primary care, and leveraging telehealth to improve access among underserved populations. The declining national prevalence of bipolar disorder warrants further investigation to determine whether this reflects true epidemiological change, improved management, or potential underdiagnosis. In contrast, the sharp rise in ADHD prevalence calls for exploration of potential contributing factors such as increased awareness, evolving diagnostic practices, and environmental or behavioral influences. Future research should prioritize longitudinal designs to disentangle causal pathways and evaluate intervention impacts as well as cross-national comparisons to identify cultural and systematic factors influencing prevalence patterns.

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Data Availability The datasets used in this study are publicly available from the Substance Abuse and Mental Health Services Administration (SAMHSA) website at https://www.samhsa.gov/data/data-we-collect/mh-cld-mental-health-client-level-data.

Declarations

Conflict of interest The authors declare no conflicts of interest.

Ethics Statement This study used publicly available, de-identified Mental Health Client-Level Data (MH-CLD) provided by the Substance Abuse and Mental Health Services Administration. Because the dataset contains no identifiable information, the research was exempt from institutional review board approval and informed consent requirements.

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